**18** A high speed train travels a distance of 487 km in 3 hours.

The distance is measured correct to the nearest kilometre.
The time is measured correct to the nearest minute.

By considering bounds, work out the average speed, in km/minute, of the train to a suitable degree of accuracy.
You must show all your working and give a reason for your answer.

3 hours = 180 minutes

$179.5 \leq t < 180.5$ minutes

**1 mark**
Half a minute each way.

$486.5 \leq d < 487.5$ km

**1 mark**
Half a kilometre each way.

Lower speed: least distance / greatest time

| | Minutes | km | |
|---|---|---|---|
| ÷180.5 | 180.5 | 486.5 | ÷180.5 |
| | 1 | $\frac{973}{361}$ | |

**1 mark**
Use opposite bounds for the speed ratio.

Lower bound = 486.5 ÷ 180.5 = 2.695290859...

Upper speed: greatest distance / least time

| | Minutes | km | |
|---|---|---|---|
| ÷179.5 | 179.5 | 487.5 | ÷179.5 |
| | 1 | $\frac{975}{359}$ | |

**1 mark**
Both bounds from correct working.

Upper bound = 487.5 ÷ 179.5 = 2.715877437...

Both round to 2.7 km/min (1 d.p.); 2 d.p. is not reliable.

**1 mark**
Both bounds round to 2.7.

2.7 km/minute

**(Total for Question 18 is 5 marks)**